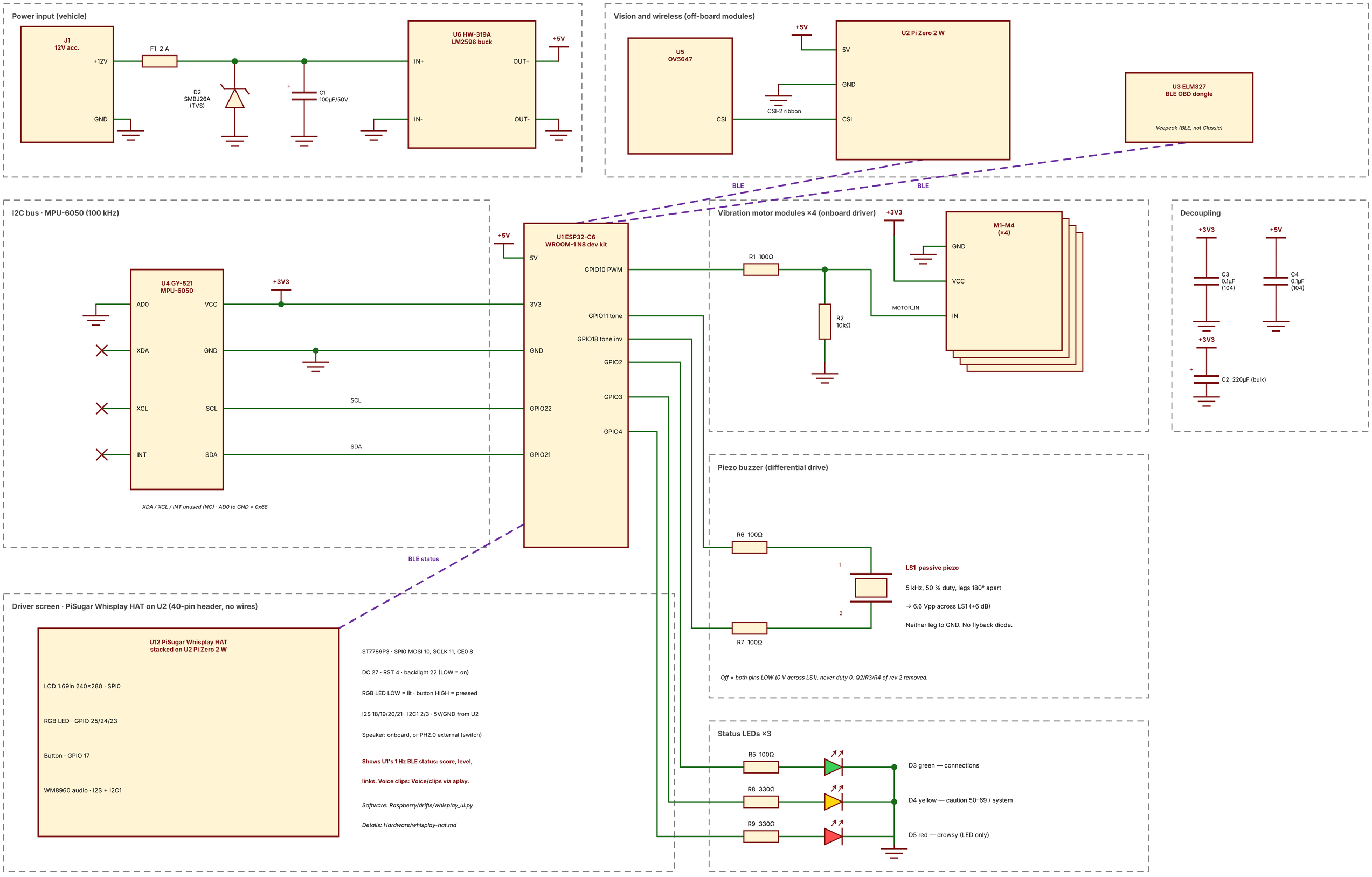
DRIFTS — Schematic (NewBrain architecture) · rev 6

Team 1 · ESP32-C6 dev kit + Raspberry Pi Zero 2 W · rev 6: driver screen, RGB LED, voice and button on the PiSugar Whisplay HAT (Pi Zero) · ESP32 keeps buzzer, motors, 3 LEDs, MPU-6050 · generated drawing (not a KiCad file)



NOTES

1. U6 = HW-319A-V7.0.1 LM2596 module. Trim the output to 5.0 V with the trimmer BEFORE connecting the Pi or C6. Bench/demo: skip the 12 V front end and use a USB power bank.
2. Motor modules M1–M4 (GND, VCC, IN) have an onboard driver. IN ← GPIO10 (20 kHz PWM, 8-bit) via R1 100Ω; R2 10k holds IN low during reset.
3. Module VCC drawn from +3V3. Verify the module rating and that the C6 LDO can supply 4 × 60–100 mA plus the IMU; +5V is the alternative.
4. U4 GY-521: AD0→GND = 0x68. Wire.begin(21, 22). Breakout has pull-ups.
5. I2C runs at 100 kHz and carries only U4 (3.3 V: no level shifter). I2C SCAN lists 0x68. Free C6 pins: 19, 20, 23.
6. LEDs: D3 green GPIO2 R5 (100Ω true green, 330Ω yellow-green), D4 yellow GPIO3 R8, D5 red GPIO4 R9. GPIO4 is a strapping pin: an LED to GND only, nothing else. Mount the LEDs in the passenger's view.
7. LS1 passive piezo, differential: GPIO11→R6→LS1←R7←GPIO18, one LEDC channel, GPIO18 inverted; 5 kHz 50 %, 6.6 Vpp. Off = both pins LOW.
8. C3/C4: 0.1µF ceramic at +3V3 and +5V near U1; one per module VCC.
9. Wireless (dashed purple): Pi↔C6 and OBD dongle↔C6 over BLE. U3 must be BLE, not Classic. Camera U5 connects to the Pi only.
10. Do not use: 5, 8, 9, 15 (strapping); 6, 7 (JTAG); 12, 13 (USB); 16, 17 (UART0 to CH343); 14, 24–30 (flash). GPIO4 = red LED only.
11. Same-named flags (+5V, +3V3, GND) are one net, as in KiCad.
12. Firmware v19: no display on the C6; its 1 Hz status JSON goes to every BLE subscriber (phone and U2). See Hardware/pinout.md.
13. U12 Whisplay HAT: screen, RGB LED, speaker, button, driven by U2. The alert (buzzer, motors, LEDs) stays on U1 and works without U2.
14. U2 + U12 power: 5 V 2.5 A on the bench; the U6 buck in the car.